Shripal Jawahar Shah

shahshripal.j@gmail.com | +1 (510) 241-8405 | San Jose, CA | LinkedIn | GitHub

Summary

Data Engineer with 4+ years of experience in building scalable ETL pipelines, data lakes, and cloud-based architectures. Proficient in SQL, Python, and Apache Spark for distributed data processing, leveraging Hadoop, Airflow, and Kafka for workflow orchestration and real-time streaming. Skilled in data modeling, data governance, and CI/CD automation using Jenkins, Docker, and Kubernetes. Experienced in machine learning (ML), NLP, and deep learning for predictive analytics. Strong expertise in AWS, Azure, and GCP cloud services, optimizing data warehousing, analytics, and visualization with Power BI and Tableau.

Technical Skills

- **Programming Languages:** Python, SQL, R, C/C++, JavaScript
- **Frameworks & Tools:** NumPy, Pandas, Matplotlib, Seaborn, Flask, Django, Git, GitHub, Docker, Jenkins, SAS, Hadoop, Hive, PySpark, ggplot2, SparkSQL, Azure Cloud, AWS Cloud, Apache Airflow, Apache Kafka, Apache Spark, ETL, SSIS, SSRS, SSAS, Talend, Agile, Statistics
- **Databases:** MySQL, Snowflake, BigQuery, PostgreSQL, MongoDB, Oracle, NoSQL, Firestore
- **Data Analysis / ETL:** Power BI, Tableau, Looker, Databricks, MS Excel, Google Analytics, Data Modeling, Data Mapping, Data Mining, Data Extraction, Transformation, Informatica, Data Lakes, Data Warehousing, Metadata Management
- **Machine Learning:** Regression, Classification, Clustering, Random Forest, Naïve Bayes, Gradient Boosting, NLP, Large Language Models (LLMs), Transformers, Computer Vision, Hugging Face, Generative AI (GenAI)

Professional Experience

Data Engineer, *IBM* 09/2024 – Present | Remote, USA

- Implemented a customer churn prediction model using Python, Scikit-learn, and TensorFlow, improving marketing efforts by identifying high-risk customers.
- Developed a real-time data pipeline using Apache Kafka and Databricks to process daily customer interaction data, reducing latency by 40% and improving data accessibility for analytics teams.
- Automated ETL workflows with Apache Airflow, integrating data from multiple sources into Snowflake and Databricks, increasing data availability and reducing processing time.
- Deployed a model on AWS SageMaker, leveraging S3 and EC2 to scale training and inference processes, reducing infrastructure costs by 20% through efficient resource allocation.
- Built an NLP-powered sentiment analysis pipeline using Hugging Face Transformers to detect customer dissatisfaction trends and enhance customer experience strategies.
- Optimized scalable data transformations using PySpark on Databricks, improving query performance and reducing data processing time by 35%, enhancing reporting and business intelligence capabilities.

Data Engineer, *GrantAide* 05/2024 – 08/2024 | San Francisco, USA

- Developed and launched an AI-driven grant-writing platform as a Minimum Viable Product (MVP), leveraging LLMs to automate funding applications, making the process faster and more efficient.
- Built a scalable data pipeline with Apache Spark and Databricks to process 20+ GB of grant data, reducing preparation time by 40% and improving data quality.
- Architected and implemented a Retrieval-Augmented Generation (RAG) pipeline leveraging LangChain and FAISS, integrated with Google Vertex AI to index 100,000+ grant documents, reducing manual research time by 90% while enhancing response accuracy.
- Optimized AI model performance by accelerating data retrieval with Python-based Cloud Run functions, improving response times by 30% for faster results.
- Developed RESTful APIs with Flask to connect AI models with Firebase databases, reducing processing time by 30% and ensuring real-time access to grant data.
- Configured a Cloud Firestore database for real-time storage of user queries and grant-related answers, improving data retrieval and response accuracy by 60% through advanced analytics and dynamic updates.

Data Engineer, *Vihas Technology LLP* 03/2019 – 11/2021 | Karnataka, India

- Designed and implemented a scalable ETL pipeline using Python, Apache Airflow, and SQL, automating financial data processing from multiple sources, reducing processing time by 30%, and enhancing fraud detection and risk analytics.
- Optimized SQL queries in PostgreSQL and MySQL, reducing execution time by 40% to improve real-time insights into customer spending patterns and enable personalized financial product recommendations.

- Developed predictive models using time series forecasting and regression to analyze stock market trends and improve credit risk assessment, leading to a 10% increase in customer retention through personalized investment insights.
- Conducted A/B testing leveraging Python (Pandas, SciPy, NumPy) to optimize financial product recommendations and UI/UX, resulting in a 10% increase in conversion rates for wealth management services.
- Built interactive Power BI dashboards to track key financial KPIs, reducing manual reporting by 50% and enhancing data-driven decision-making across risk and compliance teams.
- Implemented Git-based workflows in a CI/CD environment, improving deployment efficiency and reducing production errors by 35%, while seamlessly integrating new data models into fraud detection and investment systems.

Education

Master of Science, *San Jose State University* 01/2022 - 12/2023 | California, USA
Data Analytics

Bachelor of Engineering, *Visvesvaraya Technological University* 08/2014 - 07/2018 |Karnataka, India
Computer Science and Engineering

Projects

Data Sanitization Pipeline for Fraud Detection

- Developed a data sanitization pipeline for the Credit Card Fraud Detection Dataset, utilizing Isolation Forest and Z-score methods to detect and correct mislabeled transactions, improving fraud detection accuracy by 30%, reducing false fraud alerts by 20%, and increasing accurate fraud identification by 15%.

Smart Image Store

- Forged a scalable image storage platform with Google Cloud Vision API for image tagging and classification, using a HashMap based search algorithm and MongoDB for 30% faster retrieval; designed a real-time Tableau dashboard to visualize usage trends and storage metrics, optimizing performance with AWS load balancing and auto-scaling to handle a 60% increase in user traffic.

Publication

CMI: Cluster-Centric Missing Value Imputation with Feature Consistency (Link)

- Designed the Cluster-Centric Missing Value Imputation (CMI) framework, leveraging k-means clustering and SHAP values, improving F1 scores by 15% and increasing interpretability across datasets with up to 80% missing data.
- Implemented and benchmarked CMI on healthcare datasets (ILPD and CKD), outperforming methods like KNN and MICE, resulting in 10% higher prediction accuracy and enhanced robustness in data recovery.
- Identified top predictors for liver and kidney diseases, improving predictive accuracy by 12% and aligning results with clinical research to enhance healthcare modeling.